

Q) AI-Assignment or
 Alamu Yash Raj
 Roll no: 24k-0737

Q) Q#1.

part(a):- Binary

$$x_1 \leq P \text{ or } x_2 \leq P,$$

$$x_2 \leq E \text{ or } x_3 \leq E,$$

$$x_3 \leq E \text{ or } x_4 \leq E,$$

$$x_4 \leq P \text{ or } x_5 \leq P,$$

$$x_5 \leq P \text{ or } x_6 \leq P, x_1 \leq P \text{ or } x_6 \leq P$$

$$\forall i, j \text{ s.t. Adj}(i, j) \rightarrow (x_i \leq E \wedge x_j \leq E)$$

Unary

$$x_2 \neq P, x_3 \neq P, x_4 \neq P$$

Q) formulation
part(b):-

x_1		E
x_2		
x_3		$G \ E$
x_4		$G \ E$
x_5	P	$G \ E$
x_6	P	Pr

part(c):-

x_1		E
x_2		E
x_3		$G \ E$
x_4		$G \ E$
x_5	P	
x_6	P	Pr

Names Yash Raj

Roll no: 24K-0737

part (d) :-

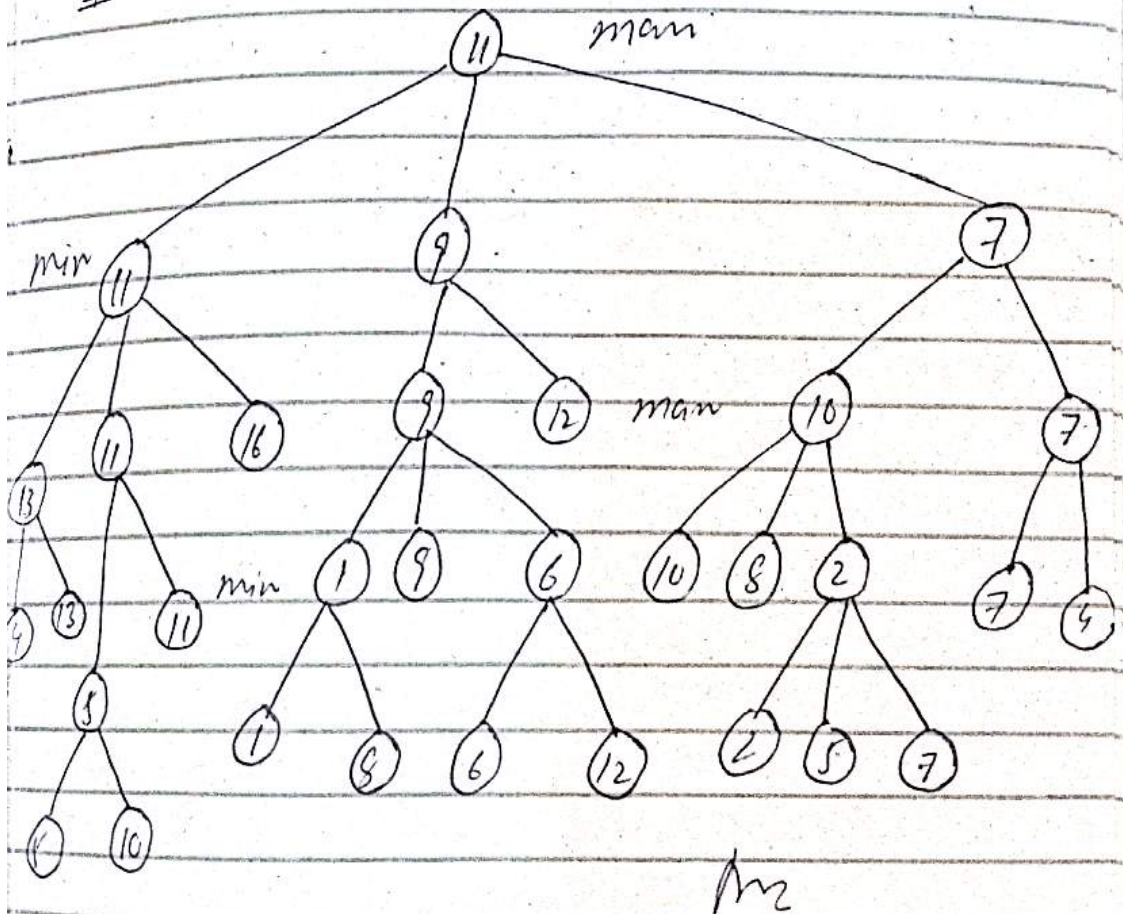
x_1	P
x_2	G E
x_3	G E
x_4	G E
x_5	P
x_6	P G E M

part (e) :-

Ans = {P, E, G, E, P, G}

{P, G, E, G, P, G}

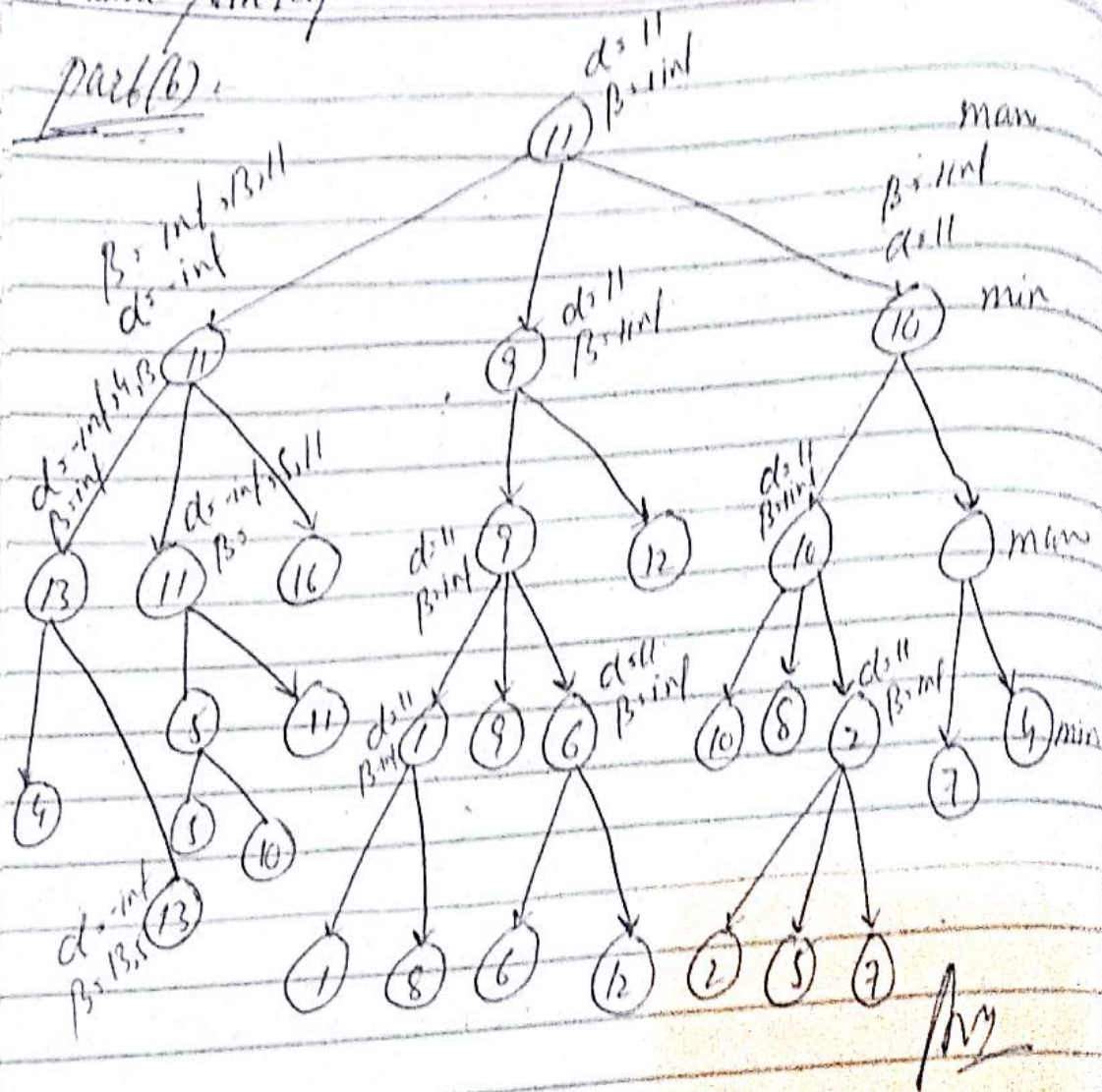
Question #2 :-
part (c) :-



Name: Yash Raj

Roll no: 24K6737

part (b):



part (c): No answer available My

Name: Yail Raj
Question #3.

Roll no: 24K-0737

$$\text{part (a)}: \frac{1}{3} \left(\frac{13}{39} + \frac{17}{51} + \frac{15}{47} \right) = 0.3286$$

$$\text{part (b)}: \frac{1/39}{\frac{1}{39} + \frac{0.42}{51} + \frac{2}{47}} = 0.3760$$

$$\text{part (c)}: \frac{17/51}{\frac{13}{39} + \frac{17}{51} + \frac{16}{47}} \approx 0.3310$$

$$\text{part (d)}: \left(\frac{1}{6} \right) \left(\frac{13}{39} \right) + \left(\frac{1}{3} \right) \left(\frac{17}{51} \right) + \left(\frac{1}{2} \right) \left(\frac{15}{47} \right) \approx 0.3262$$

$$\text{part (e)}: \frac{\left(\frac{1}{6} \right) \left(\frac{1}{39} \right)}{\left(\frac{1}{6} \right) \left(\frac{1}{39} \right) + \left(\frac{1}{3} \right) \left(\frac{0}{51} \right) + \left(\frac{1}{2} \right) \left(\frac{2}{47} \right)} \approx 0.1673$$

$$\text{part (f)}: \frac{\left(\frac{1}{3} \right) \left(\frac{17}{51} \right)}{\left(\frac{1}{6} \right) \left(\frac{13}{39} \right) + \left(\frac{1}{3} \right) \left(\frac{17}{51} \right) + \left(\frac{1}{2} \right) \left(\frac{16}{47} \right)} \approx 0.3289$$

Name: Yash Raj

Roll-no: 24K-0737

Question #4.

part(a) :- $P(R) = ?$
 $P(R/s) \Rightarrow \frac{P(R \cap s)}{P(s)}$

Sunny	Rain
S	0.25
TS	0.75

$$P(R) = \frac{P(R/s) \cdot P(s) + P(R/TS) \cdot P(TS)}{P(R)}$$

$$= \frac{(0.7)(0.25) + (0.3)(0.75)}{P(R)}$$

$$P(R) = 0.4 \text{ m}$$

part(b) :-

probability of grass is wet?
 $P(W)$

Rain	Worms	Wet
R	W	0.7 0.12
R	TS	0.25
TS	W	0.2 0.1
TS	TS	0.08

$$P(W) = P(R) \cdot P(W/R, W) + P(R) \cdot P(W/R, TS) + P(TS) \cdot P(W/TS, W) + P(TS) \cdot P(W/TS, TS)$$

$$= 0.4 \cdot 0.12 + 0.4 \cdot 0.25 + 0.6 \cdot 0.1 + 0.6 \cdot 0.08$$

$$= 0.114$$

$P(W) = 0.114$

part(c) :- $P(m/s)$ but before compute $P(w_e/s)$.

$$\begin{aligned} P(w_e/s) &= P(w_e/R, w_o) (P(w_o/R) (P(R/s))) + 0.0588 \\ &+ P(w_e/\neg R, w_o) \cdot (1 - P(w_o/R)) \cdot P(R/s) + 0.0528 \\ &+ P(w_e/\neg R, w_o) \cdot P(w_o/\neg R) \cdot (1 - P(R/s)) + 0.006 \\ &+ P(w_e/\neg R, \neg w_o) \cdot (1 - P(w_o/\neg R)) \cdot (1 - P(R/s)) + 0.0192 \end{aligned}$$

$$\begin{aligned} P(m/s) &= P(m/w_e) \cdot P(w_e/s) + P(m/\neg w_e) \cdot (1 - P(w_e/s)) + 0.1365 \\ &= (0.02)(0.1365) + 0.42(1 - 0.1365) \\ &= 0.00273 + 0.36267 \end{aligned}$$

$$\boxed{P(m/s) = 0.3654}$$

part(d) :- $P(w_o/s) = P(w_o/R) P(R/s) + P(w_o/\neg R) \cdot (1 - P(R/s))$
 $P(w_o/s) = (0.7)(0.7) + (0.2)(0.3)$
 $\boxed{P(w_o/s) = 0.55}$

$$\begin{aligned} P(B/s) &= P(B/s, w_o) (P(w_o/s)) + P(B/s, \neg w_o) \cdot (1 - P(w_o/s)) \\ P(B/s) &= 0.44 + 0.18 \\ \boxed{P(B/s) = 0.62} \end{aligned}$$

part(e) :-

$$P(w_o/s) = 0.85$$

$$P(s, \neg w_o) = 0.1125$$

$$P(s, B, \neg w_o) = (0.4)(0.1125) = 0.045$$

$$P(B, \neg w_o) = 0.24$$

$$P(s/B, \neg w_o) = \frac{P(s, B, \neg w_o)}{P(B, \neg w_o)} = \frac{0.045}{0.24} \approx 0.1875$$

$$\boxed{P(s/B, \neg w_o) \approx 0.1875}$$

Name: Yash Raj

Roll no: 24K-0783

Question #5 -

part(a) -

$$\begin{aligned}P(\text{spam}) &= 0.1 \\P(\text{study}/\text{spam}) &= 1/10 \\P(\text{study}/\text{regular}) &= 2/3 \\P(\text{free}/\text{spam}) &= 9/10 \\P(\text{free}/\text{regular}) &= 1/3 \\P(\text{money}/\text{spam}) &= 1/2 \\P(\text{money}/\text{regular}) &= 1/3\end{aligned}$$

part(b) -

$$\frac{P(\text{spam}) P(\text{study}/\text{spam}) (1 - P(\text{study}/\text{spam}))}{P(\text{money}/\text{spam})}$$

$$P(\text{spam}/i) = \frac{P(\text{spam}) (P(\text{study}/\text{spam}) (1 - P(\text{study}/\text{spam})) (P(\text{money}/\text{spam})))}{P(\text{spam}) \cdot P(\text{study}/\text{spam}) \cdot P(1 - P(\text{free}/\text{spam})) \cdot P(\text{money}/\text{spam})}$$

$$\frac{1}{P(\text{regular}) \cdot P(\text{study}/\text{regular}) (1 - P(\text{study}/\text{regular}))}$$

$$P(\text{money}/\text{spam})$$

$$P(\text{spam}/i) = 0.003736$$

part(c) -

$$0.5 = P(\text{spam})$$

$$\left[\frac{4/27 - P(\text{spam})}{5/60} \right] = 773$$

$$= P(\text{spam})/100 \text{ iff } P(\text{spam}) = 0.9674$$

Namansh Ash-Raj
Question #16:

Roll no: 21K0737

Q) What is the full equation for the regression line that you calculated?

x^2	y^2	$x \cdot y$
4	18769	274
16	4900	280
4	33856	368
25	0	0
16	1225	140
0	88209	0
9	14884	366
25	1	5
1	64009	253
9	22500	450

$$m = \frac{n(\sum xy) - (\sum x)(\sum y)}{n(\sum x^2) - (\sum x)^2}$$

$$b(\text{intercept}) = \frac{\sum y - m(\sum x)}{n}$$

$$\sum x = 2 + 4 + 2 + 5 + 4 + 0 + 3 + 5 + 1 + 3 = 29$$

$$\sum y = 137 + 70 + 184 + 0 + 35 + 297 + 122 + 1 + 253 + 150 = 1249$$

$$\sum x^2 = 4 + 16 + 4 + 25 + 16 + 0 + 9 + 25 + 1 + 9 = 109$$

$$\sum y^2 = 18769 + 4900 + 33856 + 0 + 1225 + 88209 + 14884 + 1 + 64009 + 22500 = 248353$$

$$\sum xy = 274 + 280 + 368 + 0 + 140 + 0 + 366 + 5 + 253 + 450 = 2136$$

$$\sum xy = 274 + 280 + 368 + 0 + 140 + 0 + 366 + 5 + 253 + 450 = 2136$$

$$\sum x = 29$$

$$\sum y = 1249$$

$$\sum x^2 = 109$$

$$\sum y^2 = 248353$$

$$\sum xy = 2136$$

$$m = \frac{10(2136) - (29)(1249)}{10(109) - 109}$$

$$m = \frac{-14861}{-109} = -59.683$$

$$b = 297.98$$

$$y = mx + b$$

$$y = -59.683x + 297.98$$

$$y = -59.683x + 297.98$$

Name Yash Raj

Roll.no: 24K-0733

Question #7:-

Logistic Regression Model:-

$$P(Y=1) = \frac{1}{1 + e^{-z}}$$

$$z = (b_0 + b_1x_1 + b_2x_2 + b_3x_3 + b_4x_4 + b_5x_5)$$

x_1 Age

x_2 Diabetic

x_3 Smoker Hypertension

x_4 Smoker

x_5 Surgery Type

b_0 is intercept

Variable	Coefficient
b_0	-8

Age (b_1)

0.05

Diabetic (b_2)

1.5

Hypertension (b_3)

1.2

Smoker (b_4)

1.0

Surgery Type (b_5)

2.0

2) Compute linear

combination (z).

$$z = (-8) + (0.05)(60) + (1.5)(1) + (1.2)(1) + (1.0)(1) + (2.0)(2)$$

$$z = -8 + 3 + 1.5 + 1.2 + 1 + 4$$

$$z = -8 + 3 + 1.5 + 1.2 + 1 + 4$$

$$[z = 2.7] \text{ Ans}$$

2) Patient data:-

feature	Value
Age	60
Diabetic	1
Hypertension	1
Smoker	1
Surgery Type	2

2) probability:-

$$P(Y=1) = \frac{1}{1 + e^{-z}}$$

$$e^{-2.7} = 0.067$$

$$\frac{1}{1 + 0.067}$$

$$[P(Y=1) = 0.937]$$

Ans